CHAPTER EIGHT

Processing serial synchrotron crystallography diffraction data with DIALS

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Abstract

This chapter describes additions to the DIALS software package for processing serial still-shot crystallographic data, and the implementation of a pipeline, *xia2.ssx*, for

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processing and merging serial crystallography data using DIALS programs. To integrate partial still-shot diffraction data, a 3D gaussian profile model was developed that can describe anisotropic spot shapes. This model is optimised by maximum likelihood methods using the pixel-intensity distributions of strong diffraction spots, enabling simultaneous refinement of the profile model and Ewald-sphere offsets. We demonstrate the processing of an example SSX dataset where the improved partiality estimates lead to better model statistics compared with post-refined isotropic models. We also demonstrate some of the workflows available for merging SSX data, including processing time/dose resolved data series, where data can be separated at the point of merging after scaling and discuss the program outputs used to investigate the data throughout the pipeline.

1. Introduction

Macromolecular X-ray crystallography has been a very successful technique for determining biological molecular structures, contributing to over 180,000 entries in the protein data bank (PDB) to date. The typical practice of determining structures at cryogenic temperatures, while justified in order to reduce radiation damage and enable complete data collection from a single crystal by rotation, is less comparable to native biological conditions than data collection at room temperature. This is of particular importance for the study of dynamics in structural biology by crystallography, a field that has had a renaissance over the past decade, enabled by several experimental developments in X-ray crystallography (Orville, 2018). This includes the development of sample delivery methods such as fixed-targets, liquid and viscous jets to rapidly present thousands of microcrystals for measurement (Grünbein & Nass Kovacs, 2019), in combination with the development of time-resolved serial femtosecond crystallography at X-ray free-electron lasers (XFELs) (Pandey et al., 2020), and synchrotron advancements including increased photon flux density and the development of micro-focus beamlines (Axford et al., 2012). The use of both XFEL and synchrotron facilities therefore enables biological processes to be investigated on a wide range of timescales at room temperature, down to the femtosecond timescale at XFELs or typically on the milli- or micro-second timescale at synchrotrons. At XFELs, the impact of radiation damage is mitigated by the extremely fast timescales and high X-ray intensities, which produce strong diffraction on a single image before sample destruction (Neutze et al., 2000) or site-specific radiation damage, if a suitably short pulse is used (Hough & Owen, 2021). At synchrotrons, collecting only a few images (or a single still image) from any single crystal

limits the radiation dose, which is calculable and controllable. In both cases, a complete dataset of sufficient data quality can only be achieved by combining data from hundreds or thousands of crystals.

From a data processing perspective, these experimental developments present new challenges. These techniques are capable of measuring diffraction from over hundreds of crystals a second, however the number of images that need to be collected cannot be specified ahead of the experiment due to a number of factors. Firstly, not every image will contain diffraction from a crystal, due to the nature of the rapid sample delivery techniques. Secondly, the overall merged data quality (resolution, completeness, internal consistency) from a large number of crystals will determine whether biological insight can be attained, therefore enough data must be collected, but over-collecting will reduce the number of different experiments that can be performed in the allocated experiment time. This means that fast feedback is critical to a successful experiment and making efficient use of available beamtime; in its simplest form this can be live hit-finding analysis to give an indication of the number of crystals exposed, but as each step of data analysis provides more information, ideally the complete merged data would be returned as soon as possible (i.e. within a matter of minutes). Examples of software that provide online data analysis at XFELs include Cheetah (Barty et al., 2014), OnDA (Mariani et al., 2016), IOTA (Lyubimov et al., 2016), which themselves run analysis algorithms contained in packages such as CrystFEL (White et al., 2012), DIALS (Winter et al., 2018), cctbx.xfel.merge (Brewster et al., 2019a) and XDS (Kabsch, 2010b). In addition, live analysis may be performed as part of specialised hardware-accelerated processing, for example in the Jungfraujoch system (Leonarski et al., 2023).

A second challenge is that processing still-shot diffraction data presents algorithmic challenges compared with rotation data. For rotation data, each reflection is rotated through the diffraction condition, so the measured intensity is the integral of the complete diffraction spot and the profile (shape) of the spot can be empirically determined for suitably fine-sliced data. For still-shot data, each diffraction pattern only measures a slice through the volume of the diffraction spots. In addition, most reflections will not be measured at the exact diffraction condition for the ideal lattice, therefore the diffraction profile must be estimated to model these offsets in order to determine an accurate partiality estimate (sometimes referred to as the Ewald offset correction) that puts the measured intensities on the same scale within an image. One method to address this for serial crystallography

is the post-refinement method, where following integration of reflection intensities on the images, the crystal parameters such as unit cell, orientation and mosaic spread, which together determine the partiality, are refined alongside the scaling corrections to give the most internally consistent merged intensities (Uervirojnangkoorn et al., 2015). However, post-refinement approaches are constrained by the fidelity of the models employed, which typically apply isotropic models of mosaic block size and spread (Kabsch, 2014; Sauter et al., 2014), plus a polychromatic beam-model when modelling XFEL data (White, 2019; Ginn et al., 2016). Work by Mendez et al. (2020) showed how a full experiment model can be refined against individual pixel data using maximum likelihood methods, leading to improved data quality but relying on significant computational resources that would be impractical for processing during experimental data collection. Recent work by Brehm et al. (2023) describe a more general gaussian model for describing anisotropic peak shapes that leads to improved data quality, a similar approach to the integration algorithm that we describe here, first described in Parkhurst (2020). Another algorithmic challenge when combining thousands of measurements is the indexing ambiguity problem for polar space groups, whereby all images must be consistently indexed prior to scaling and merging, a particular challenge in serial crystallography due to the sparsity of common reflections between still images. Algorithms for indexing ambiguity resolution of rotation data have been described for rotation data (Brehm & Diederichs, 2014; Gildea & Winter, 2018) and have implementations in XDS, DIALS and CrystFEL.

In this work we describe and evaluate recent additions and developments in the DIALS software package for the processing of still-shot serial synchrotron crystallography (SSX) data. This includes a description of an enhanced profile modelling algorithm for integration of still images, using a 3D gaussian description to model anisotropic spot profiles and reflection offsets based on pixel intensity distributions (Parkhurst, 2020) now implemented in DIALS for a monochromatic beam model. Using SSX data collected on beamline I24 at Diamond Light Source, we show how this leads to improved partiality estimates compared with post-refined isotropic partiality models, with improved structure quality for a given number of crystals, or requiring significantly fewer crystals to reach a given overall *R*-factor. In addition, we describe an automated processing pipeline for serial data processing, *xia2.ssx*, running stepwise processes from DIALS that enables full processing from images to merged data, fulfilling an equivalent

role to the *xia2* (Winter, 2010) pipeline for processing crystallographic rotation data. The *xia2.ssx* pipeline includes additional workflows required for serial crystallography, such as separation of data by experimental criteria, for example in dose or times series experiments, as well as quick merging of integrated data during data accumulation in an SSX experiment. Given the diverse nature of serial crystallography experiments, there is no single workflow for all experiments, and therefore the focus here is on demonstrating the features that are available for processing and the program outputs that can be used for evaluating the data processing. These developments in DIALS and *xia2* provide a consistent data processing experience at synchrotrons from rotation to serial crystallography, lowering the barrier to entry for experimentalists experienced in rotation crystallography processing methods.

2. Methods

Crystalline diffraction produces observed diffraction spots that have a spatial spread on the detector surface, since a crystal does not form a perfect infinite lattice and the X-ray beam has a finite spatial divergence and wavelength distribution. In the reciprocal space formalism, there is a distribution of intensity about the ideal reciprocal lattice points (RLPs), while the diffraction condition as described by the Ewald sphere is broadened by the divergence and wavelength distribution of the beam.

During a rotation experiment, the entire reflection is rotated through the diffracting condition, such that the intensity weighted centre of mass of the observed reflection on the detector image corresponds to the predicted position of the reflection, for a suitably well modelled crystal. For still-shot diffraction data, the intensity distribution of an observed spot on the detector is the result of the convolution of the Ewald Sphere distribution with the broadened reciprocal lattice point. As such, the weighted centre of mass of the observed reflection, when mapped into reciprocal space, will not lie on the centre of the reciprocal lattice point, with the deviation depending on the shape of the reciprocal lattice points and the orientation of the crystal, thus varying from reflection to reflection. To correct for this effect, a per-reflection scale factor is required, which has been termed an Ewald offset correction or partiality in previous literature (Kabsch, 2014; White et al., 2013). The term 'partiality' will be used in this work as a shorthand for this geometry dependent scale factor, however this is distinct

from the definition of partiality of a reflection in a rotation experiment, as the full reflection intensity is not measured in a still-shot image.

Within DIALS, the existing stills profile model consists of two components (Sauter et al., 2014); the mosaic block size which results in spherical broadening of RLPs, and the angular spread of mosaic blocks, which results in a 'spherical cap' around the reciprocal lattice vector (Nave, 1998). For these spherical (isotropic) models, transformations can be applied to determine the predicted centres of mass of the spots on the detector from the models (Sauter et al., 2014).

In this section, we describe an approach using a multivariate normal distribution (MVN) to describe anisotropic RLPs. For non-isotropic RLPs, the centre of mass offset is highly dependent on the shape of the profile. The use of a MVN description combined with a maximum likelihood optimisation leads to mathematical simplifications and allows simultaneous modelling of the Ewald offsets and RLP shapes with a per-image model of unit cell, crystal orientation and anisotropic mosaic spread.

2.1 Model parameterisation using a multivariate-normal distribution

The distribution of RLPs in reciprocal space, can be modelled as a multivariate normal distribution convolved with the reciprocal lattice and described by a 3D covariance matrix $\mathbf{M}$. A valid covariance matrix is required to be positive semi-definitive, which is enforced by using a Cholesky decomposition, i.e. $\mathbf{M} = \mathbf{L}\mathbf{L}^*$, where $\mathbf{L}$ is a lower triangular matrix with positive diagonal elements. Therefore, the RLP covariance matrix is fully described by 6 parameters which are the non-zero elements of $\mathbf{L}$, where

$$\mathbf{L} = \begin{pmatrix} m_1 & 0 & 0 \\ m_2 & m_3 & 0 \\ m_4 & m_5 & m_6 \end{pmatrix} \tag{1}$$

In the reciprocal-space coordinate frame, this describes an arbitrarily orientated ellipsoidal distribution. Spherical spot shapes can be modelled with a constrained form of $\mathbf{L}$ with $m_1 = m_3 = m_6 > 0$, $m_2 = m_4 = m_5 = 0$. In these models, which will be referred to as $\mathbf{M_E}$ (ellipsoidal) and $\mathbf{M_S}$ (spherical), all RLPs have an identical covariance matrix. Fig. 1A shows example RLP distributions for an ellipsoidal covariance matrix.

To model an angular mosaic spread, a radial dependence must be introduced to the per-reflection RLP distribution to describe RLPs that subtend a constant angle with the reciprocal space origin, which can model

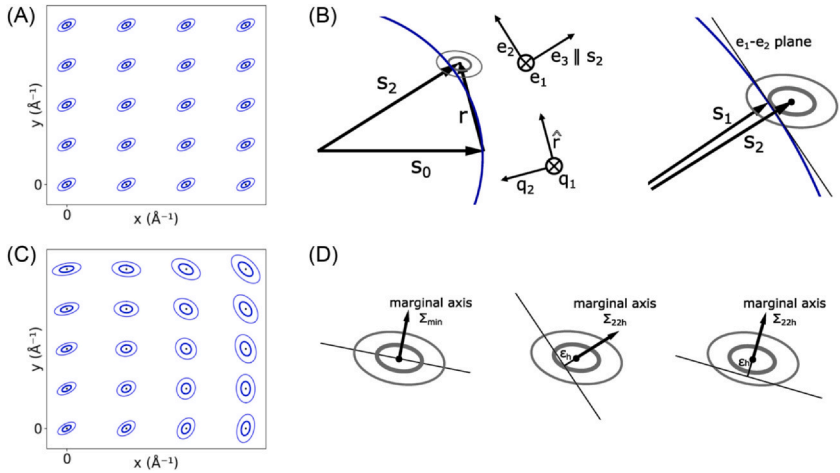

Fig. 1 (A) Example reciprocal lattice point (RLP) distributions described by a reflection independent ellipsoidal covariance matrix $\mathbf{M_E}$, plotted for contours of fixed probability density. (B) Schematic of vector and coordinate frame definitions. $\mathbf{s_2}$ is the vector to the RLP centre and $\mathbf{r} = \mathbf{s_2} - \mathbf{s_0}$. In general, $|\mathbf{s_2}| \neq |\mathbf{s_0}|$ and the Ewald sphere does not intersect at the RLP centre. The $\mathbf{e}$ vectors define the orthonormal local reflection-specific coordinate frame, which is used to separate the multivariate gaussian into a conditional distribution in the $\mathbf{e_1}$-$\mathbf{e_2}$ plane and a marginal distribution along $\mathbf{e_3} \parallel \mathbf{s_2}$. The vectors $\mathbf{q_1}$, $\mathbf{q_2}$ and $\hat{\boldsymbol{r}}$ form an orthonormal set that is used to rotate an angular component of mosaic spread into the reciprocal space coordinate frame for a given reflection, with the angular component in the $\mathbf{q_1}$-$\mathbf{q_2}$ plane. $\mathbf{s_1}$, the observed centre of mass of the diffraction spot, is approximated by transforming the mean of the conditional distribution to the laboratory frame (see text). (C) Example RLP distributions described by covariance matrices $\mathbf{M^h}$ adding an angular mosaic spread to $\mathbf{M_E}$, plotted for contours of fixed probability density (relative to the probability density at the centre of each RLP). (D) Illustration of quantities used in calculation of the partiality of a spot, which depends on the distance from the Ewald sphere, ϵ, along the marginal axis, and on the variance along this axis Σ_{22h}. The highest integrated probability density occurs in the first case, for an intersection at the RLP centre perpendicular to the direction of minimum variance (Σ_{min}).

an angular distribution of mosaic domain orientations. First, a decomposition matrix $\mathbf{L_A}$ is defined, where

$$\mathbf{L_A} = \begin{pmatrix} m_1 & 0 & 0 \\ m_2 & m_3 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{2}$$

which parameterises a 2D MVN in the xy plane. The angular covariance $\mathbf{M_A^h}$ of reflection $\mathbf{h}$ is scaled in proportion to $|\mathbf{r}|$, with $\mathbf{M_A^h} = (|\mathbf{r^h}|\mathbf{L_A})(|\mathbf{r^h}|\mathbf{L_A})^*$.

Finally, a transformation matrix $\mathbf{Q^h}$ is defined to rotate the covariance into the reciprocal space coordinate frame for each reflection, where

$$\mathbf{Q^h} = \begin{pmatrix} \hat{\mathbf{q}}_1 \\ \hat{\mathbf{q}}_2 \\ \hat{\mathbf{r}} \end{pmatrix} \tag{3}$$

$\mathbf{q_1} = \hat{\boldsymbol{r}} \times \hat{\boldsymbol{s}}_0$ and $\mathbf{q_2} = \hat{\boldsymbol{r}} \times \hat{\boldsymbol{q}}_1$ (illustrated in Fig. 1B). Therefore, the angular covariance for a reflection $\mathbf{h}$ is described by the $|\mathbf{r}|$-dependent covariance matrix $\mathbf{M^h} = (\mathbf{Q^h})^{\mathbf{T}}\mathbf{M^h_A}\mathbf{Q^h}$, a 2D MVN in the $\mathbf{q_1}$-$\mathbf{q_2}$ plane perpendicular to $\mathbf{r}$.

The coordinate frames are shown in Fig. 1A. For this angular component, the 2D MVN can be constrained to be isotropic, i.e. circular, rather than elliptical, with the constraints $m_1 = m_3 > 0$, $m_2 = 0$. Purely angular mosaic spread models are known to inadequately describe observed low resolution reflections, as the angular component of covariance tends to zero as $|\mathbf{r}| \rightarrow 0$ (Sauter et al., 2014). Therefore, an additional $\mathbf{r}$-invariant mosaicity component should be included, to account for the finite size of the RLPs at low resolution, which can be achieved by including the $|\mathbf{r}|$-invariant ellipsoidal/spherical mosaicity model. An overall model then defines the reflection covariance most generally as $\mathbf{M^h} = (\mathbf{Q^h})^{\mathbf{T}}\mathbf{M^h_A}\mathbf{Q^h} + \mathbf{M_E}$: this describes a reflection-independent 3D MVN distribution of the RLPs due to domain size effects, with 2D angular broadening from orientational mosaic spread. Example RLP distributions of $\mathbf{M^h}$ are shown in Fig. 1C.

2.2 General impact of the RLP distribution

To optimise the model, the relationship between the mosaicity model parameterisation and the expected distribution of diffracted beam vectors on the Ewald sphere (referred to as the general impact) must be known. This distribution can be described simply, whereas the projection onto a flat detector plane is more complex. Under the assumption that the Ewald sphere is flat on the scale of a spot, the general impact can be approximated by the shape of the spot on a plane tangential to the Ewald sphere surface. To work in this reference frame, it is convenient to define a reflection specific coordinate frame with basis vectors ($\hat{\boldsymbol{e}}_1$, $\hat{\boldsymbol{e}}_2$, $\hat{\boldsymbol{e}}_3$), where $\mathbf{e_1} = \hat{\boldsymbol{s}}_2 \times \hat{\boldsymbol{s}}_0$, $\mathbf{e_2} = \hat{\boldsymbol{s}}_2 \times \hat{\boldsymbol{e}}_1$, $\hat{\boldsymbol{e}}_3 = \hat{\boldsymbol{s}}_2$, i.e. $\hat{\boldsymbol{e}}_1$ and $\hat{\boldsymbol{e}}_2$ define a tangent plane to the Ewald sphere (Fig. 1B). The definitions of $\mathbf{e_1}$ and $\mathbf{e_2}$ are the same as in the coordinate system defined in Kabsch (2010a), but the $\hat{\boldsymbol{e}}_3$ definition differs. These basis vectors can be used to form the rows of a rotation matrix $\mathbf{R_e} = (\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2, \hat{\boldsymbol{e}}_3)$,

which transforms the laboratory space coordinate system for a given reflection such that the $\mathbf{s_2}$ vector lies along the z-axis of the local reflection coordinate system. Therefore, given a covariance matrix $\mathbf{M}$, the RLP distribution in the local coordinate system is a MVN $P(\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2, \hat{\boldsymbol{e}}_3)$ with mean $\boldsymbol{\mu} = \mathbf{R_e s_2}$ and a covariance $\boldsymbol{\Sigma} = \mathbf{R_e M R_e^T}$. To compute the general impact requires calculating the conditional distribution of the MVN on the tangent plane, i.e. it requires the decomposition of P into the 1D marginal distribution $P(\hat{\boldsymbol{e}}_3)$ and the 2D conditional distribution $P(\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2 | \hat{\boldsymbol{e}}_3)$, with $P(\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2, \hat{\boldsymbol{e}}_3) = P(\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2 | \hat{\boldsymbol{e}}_3)\ P(\hat{\boldsymbol{e}}_3)$. A convenient property of the MVN distribution is that the marginal and conditional distributions are also normal distributions. First, we define the partition of the mean vector and covariance matrix as follows:

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma_{11}} & \boldsymbol{\Sigma_{12}} \\ \boldsymbol{\Sigma_{21}} & \Sigma_{22} \end{pmatrix}. \tag{4}$$

Here $\boldsymbol{\mu_1}$ and $\boldsymbol{\Sigma_{11}}$ are the 2D vector mean and covariance of the conditional distribution, and μ_2 and Σ_{22} are the scalar mean and variance of the marginal distribution. Using the properties of block matrix inversions (Appendix A), the conditional distribution at the intersection with the Ewald sphere can be shown to have a mean and variance

$$\bar{\mu} = \mu_1 + (\boldsymbol{\Sigma_{12}})(\Sigma_{22})^{-1}(|s_0| - \mu_2), \quad \bar{\boldsymbol{\Sigma}} = \boldsymbol{\Sigma_{11}} - (\boldsymbol{\Sigma_{12}})(\Sigma_{22})^{-1}(\boldsymbol{\Sigma_{21}}) \tag{5}$$

The mean of the conditional distribution describes the central impact of the spot on the Ewald sphere, and the variance describes the general impact of the spot. The centre of mass of the reflection recorded on the Ewald sphere can be approximated by transforming the mean of the conditional distribution back into the laboratory frame:

$$\mathbf{s_1} = \mathbf{R_e^T} \begin{pmatrix} \bar{\boldsymbol{\mu}} \\ |\mathbf{s_0}| \end{pmatrix} \tag{6}$$

2.3 Parameter estimation via maximum likelihood methods

The starting point for model optimisation is the pixel intensity values of a set of indexed spots and an experimental model (beam, detector, crystal). The pixel intensity values constitute the general impact of a spot on the detector, which are mapped into the local reflection-specific coordinate frame. In the local coordinate frame, this distribution can be approximated as a bivariate normal distribution, whereas the projection onto the detector

surface is in general not normal. This transformation and approximation significantly simplify the estimation of the model parameters and allows the parameters to be estimated via a maximum likelihood estimator. The probability of observing a count at a position x, y, z in the reflection-specific coordinate system is given by $P(x, y, z) = P(x, y|z)P(z)$. Each RLP centre is predicted to be a distance $\epsilon = |\mathbf{s_2}| - |\mathbf{s_0}|$ from the.

Ewald sphere along $\mathbf{e_3}$, and the centre of each pixel in the observed spot maps to a point $\mathbf{x} = (x, y)$ in the $(\mathbf{e_1}, \mathbf{e_2})$ plane. Given the expected mean and covariance from the model, the probability density of observing a count at $(\mathbf{x}, \epsilon)$ is

$$P(\mathbf{x} \mid \epsilon) = \frac{1}{\sqrt{(2\pi)^2 \, |\bar{\mathbf{\Sigma}}|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \bar{\boldsymbol{\mu}})^{\mathbf{T}} \bar{\mathbf{\Sigma}}^{-\mathbf{1}} (\mathbf{x} - \bar{\boldsymbol{\mu}})\right) \tag{7}$$

$$P(\epsilon) = \frac{1}{\sqrt{(2\pi)\Sigma_{22}}} \exp\left(-\frac{1}{2}\epsilon^2 \Sigma_{22}^{-1}\right) \tag{8}$$

Therefore, given N observed strong spots, where a spot h has contributions from O_h detector pixels, j, and each pixel has observed counts of $c_{h,j}$, the log likelihood can be written as:

$$\begin{aligned} L = -\frac{1}{2} \sum_{h}^{N} \sum_{j}^{O_h} c_{h,j} [\ln(\Sigma_{22h}) + \epsilon_h^2/\Sigma_{22h} + \ln(|\bar{\mathbf{\Sigma}}_h|) \\ + (\mathbf{x}_{h,j} - \bar{\boldsymbol{\mu}}_h)^T \bar{\mathbf{\Sigma}}_h^{-1} (\mathbf{x}_{h,j} - \bar{\boldsymbol{\mu}}_h)] \end{aligned} \tag{9}$$

For each spot, the total observed counts, c_{tot}, mean, $\bar{\boldsymbol{x}}$ and covariance $\mathbf{S}$ of each spot are given by

$$c_{\text{tot}} = \sum_{j}^{O} c_j, \; \bar{\mathbf{x}} = \frac{1}{c_{\text{tot}}} \sum_{j}^{O} c_j \mathbf{x}_j, \quad \mathbf{S} = \frac{1}{c_{\text{tot}}} \sum_{j}^{O} c_j (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^T \tag{10}$$

Using the cyclic permutation property of the matrix trace operator, the log likelihood can be written in terms of these statistics as

$$\begin{aligned} L = -\frac{1}{2} \sum_{h}^{N} c_{\text{tot},h} [\ln(\Sigma_{22h}) + \epsilon_h^2/\Sigma_{22h} + \ln(|\bar{\mathbf{\Sigma}}_h|) + \operatorname{tr}(\bar{\mathbf{\Sigma}}_h^{-1} \mathbf{S}_h) \\ + (\bar{\mathbf{x}}_h - \bar{\boldsymbol{\mu}}_h)^T \bar{\mathbf{\Sigma}}_h^{-1} (\bar{\mathbf{x}}_h - \bar{\boldsymbol{\mu}}_h)]. \end{aligned} \tag{11}$$

This removes the inner sum over the spot pixels, and since the statistics c_{tot}, $\bar{\boldsymbol{x}}$ and $\mathbf{S}$ do not depend on the model parameters, they only need to be calculated once which therefore allows quick evaluation of the log-

likelihood at each step of optimisation. To estimate the parameters, the Fisher scoring algorithm (Fisher, 1922) is used, which only requires calculation of the first derivatives of L.

Further details of the calculations of the derivatives and Fisher information matrix can be found in Appendix B. The expected general impact depends on the mosaicity parameterisation in addition to the crystal model ($\mathbf{U}$ and $\mathbf{B}$ matrices), as the crystal model determines the predicted $\mathbf{s_2}$ and both must be optimised. The starting point for the optimisation is the crystal model from indexing and an isotropic $\mathbf{M}$ with diagonal elements σ_D^2, where σ_D is the standard deviation of the beam divergence (Kabsch, 2010b), calculated on the indexed spots. Maximum likelihood optimisation is performed on the mosaicity parameterisation, followed by the crystal model, after which the indexed spots are re-predicted and the spot statistics recalculated. By default, three of these macrocycles are performed for each indexed crystal.

2.4 Integration and partiality estimation

Following optimisation of the profile model, unit cell and orientation of the crystal using the strong spots, all reflections are predicted on the image and integrated using the procedure described in Winter et al. (2018). For each spot, a shoebox is generated and separated into foreground and background pixels. Foreground pixels are defined as those within the χ^2 quantile with probability 0.9973 (univariate 3 σ level) in the reflection-specific coordinate system. The integrated intensity is determined by summation of the foreground pixels minus an estimate of the background. To calculate the reflection dependent scale factor, i.e. the partiality, it is assumed that the Ewald sphere is flat on the scale of a RLP. For a monochromatic beam, i.e. a δ-function wavelength distribution, the Ewald sphere intersects the RLP as a plane, and the probability density of the RLP in this plane is given by the conditional distribution. As the conditional distribution is itself a multivariate normal distribution, the relative integrated probability density is therefore the probability density of the marginal distribution

$$\rho_h \propto \frac{1}{\sqrt{\Sigma_{22h}}} \exp\left(-\frac{1}{2}\epsilon_h^2/\Sigma_{22h}\right). \tag{12}$$

This depends on the shortest distance between the RLP centre and the intersection of the Ewald sphere and also on Σ_{22h}, the variance of the marginal distribution perpendicular to the Ewald sphere. Therefore, ρ_h is

largest when Σ_{22h} is the variance along the shortest elliptical axis and $\epsilon_h = 0$, i.e. the Ewald sphere intersects perpendicular to the shortest axis, which we refer to as $\Sigma_{\min}$, as illustrated in Fig. 1D. For an intersection off the centre of the RLP at an arbitrary orientation, the relative partiality is therefore given by

$$p_h = \frac{\rho_h}{\rho_{\min}} = \frac{\frac{1}{\sqrt{\Sigma_{22h}}} \exp\left(-\frac{1}{2}\epsilon_h^2 / \Sigma_{22h}\right)}{\frac{1}{\sqrt{\Sigma_{\min}}}} = \sqrt{\frac{\Sigma_{\min}}{\Sigma_{22h}}} \exp\left(-\frac{1}{2}\epsilon_h^2 / \Sigma_{22h}\right) \tag{13}$$

Importantly, for anisotropic RLPs, the partiality of each reflection depends on both the displacement of the RLP centre from the Ewald sphere and the orientation of the RLP covariance relative to the Ewald sphere for a given reflection. For mosaicity models with a reflection-independent covariance matrix, all reflection covariances have the same $\Sigma_{\min}$. For radially dependent models $\Sigma_{\min}$ is the minimum covariance of the reflection-independent part of the model. The relative uncertainty in $1/p$ can also be estimated as

$$\frac{\sigma(1/p_h)}{1/p_h} = p_h \frac{\partial(1/p_h)}{\partial \epsilon_h} \sigma(\epsilon_h) \approx \frac{\epsilon_h}{\Sigma_{22h}} \sqrt{\frac{\Sigma_{22h}}{n_{\text{obs}}}}, \tag{14}$$

where the estimate of the uncertainty in ϵ_h is given by the uncertainty of the mean of a normal distribution from n_{obs} observations of spots upon which the mosaicity model is optimised.

3. DIALS command-line tools for processing SSX data

The DIALS package contains a set of command line programs to enable sequential processing of rotation data from raw images to merged intensity files. In addition, the individual programs contain a range of algorithms and options suitable to different use cases including rotation and still-shot data processing. To facilitate the processing workflow for processing blocks of serial still-shot data, two new programs were implemented, *dials.ssx_index* and *dials.ssx_integrate*. These programs perform indexing and integration on blocks of still-shot images, using appropriate parameters for still-shot data and aggregating summary statistics into HTML reports and log outputs. This section summarises which aspects of the existing DIALS algorithms are used in these programs, and the new additions to data reduction programs to enable still-shot processing. These

components then form the basis of processing workflows for still-shot data processing, as implemented in the *xia2.ssx* pipeline which is detailed in Section 4.

3.1 Indexing with dials.ssx_index

The *dials.ssx_index* program runs the same underlying algorithms as *dials.index* on the output of *dials.find_spots*, a table of the strong spots per image. Indexing is attempted on each image that contains at least 10 strong spots, using the indexing option *stills.indexer=stills*. Typically, the *unit_cell* and *space_group* are set if known, which greatly increases the indexing success rate for still-shot data. Multiple lattices are searched for on each image, using the *FFT1D* indexing algorithm. If the *FFT1D* algorithm is unsuccessful and a unit cell is provided, indexing is also attempted with the *real_space_grid_search* algorithm (Gildea et al., 2014). Following refinement of the crystal model, with a fixed beam and detector model, the stills indexer refines two parameters describing the crystal mosaicity as a combination of a mosaic block size parameter and mosaic spread parameter, as described in Sauter et al. (2014). A HTML report is generated, containing plots of per-image quantities such as the percentage of strong spots indexed, and distributions of cell parameters. The output data files can be viewed in the *dials.image_viewer* and *dials.reciprocal_lattice_viewer*.

3.2 Integration with dials.ssx_integrate

The *dials.ssx_integrate* program runs the ellipsoid profile modelling and integration algorithm, as described in Section 2 on each indexed crystal. The default parametrisation used is the six-parameter ellipsoidal covariance matrix $\mathbf{M_E}$, with no angular component. This profile model was found to best-describe typical well-diffracting SSX datasets. Following max-likelihood optimisation of the profile and crystal model, the full set of intensities are predicted, background-subtracted intensities are determined by summation integration and partiality values are estimated, as described in Section 2.4. In *dials.ssx_integrate* this routine is selected by default with the option *algorithm=ellipsoid*. Alternatively, the two-parameter mosaicity model determined by the *stills* indexer can be used to predict the location and spatial extent of spots on the image, followed by summation integration as described in Winter et al. (2018), which is selected with the option *algorithm=stills*. A HTML report is generated, containing plots of per-image metrics including $I/\sigma(I)$ and the mosaicity model parameters, and the output can be viewed as before.

3.3 Data reduction with dials.cosym and dials.scale

For still-shot data, existing symmetry-determination algorithms are typically unreliable for determination of an unknown point group symmetry. Therefore, after integration, a particular space group must have been chosen to continue processing. However, only the point group affects indexing ambiguity resolution and scaling. For space groups where the point group symmetry is lower than the lattice symmetry, there is more than one valid way of indexing the diffraction pattern of a given crystal, however a consistent choice must be achieved across all crystals. The *dials.cosym* program can be used to resolve this indexing ambiguity for still-shot data, with a higher chance of success as the signal-to-noise ratio and resolution range of each image increases. An option was added to *dials.cosym* to provide a reference file, containing intensity information (MTZ or CIF file format), or a structural model (PDB or CIF file format) from which estimated intensities can be calculated. If a reference file is provided, *dials.cosym* selects the indexing solution which best correlates with the reference.

For scaling still-shot data, the kB scaling model in the *dials.scale* program is used, which consists of a scale factor (k) and *B*-factor for each crystal (Beilsten-Edmands et al., 2020). There are two different approaches for scaling which are fundamentally different in how they scale with the number of crystals. If a reference model is provided, each image can be scaled independently against the reference intensities. The reference intensity values are used as fixed merged intensity values $\langle I_h \rangle$ in the scaling objective function and outlier rejection formula. In this case, the scale factor determination for a given crystal is a linear least-squares problem with two free parameters, which can be quickly solved separately for each crystal and is trivially parallelizable. If no reference model is used, the best-estimate set of merged intensities $\langle I_h \rangle$ is calculated at each step of the optimisation by merging data from all datasets, and changes during the minimisation process. In this case, scaling is a non-linear least-squares problem with $2\,n$ free parameters, where n is the total number of crystals. This has a much higher computational complexity, higher memory requirements and becomes slow for many thousands of crystals. Another important aspect of data reduction is the adjustment of intensity uncertainties. In *dials.scale*, a two-parameter error model is optimised to adjust the uncertainty estimates, $(\sigma')^2 = a^2(\sigma^2 + (bI)^2)$, where a and b are the error model parameters to optimise. Importantly, for still data integrated with the

ellipsoid profile model, the uncertainties of the partiality estimates are calculated from the profile model and propagated to σ before the error model optimisation is performed. The inclusion of a partiality uncertainty is a key difference to rotation data and increases the σ of intensity observations with low partiality.

3.4 Associated tools from cctbx.xfel

The DIALS package depends on the cctbx package (Grosse-Kunstleve et al., 2002) for its core crystallographic calculations. Within cctbx there is a program for data reduction of XFEL data, *cctbx.xfel.merge*, which takes as input DIALS integrated data files integrated with the *stills* integration algorithm. As described in previous publications (Sauter, 2015; Brewster et al., 2019b), *cctbx.xfel.merge* implements a post-refinement algorithm that optimises the two-parameter mosaicity model of the *stills* indexer, the crystal orientation, a scale and *B*-factor as part of data reduction. Additionally, a three-parameter error model $(\sigma')^2 = \mathrm{SdFac}^2(\sigma^2 + (\mathrm{SdB} * I) + (\mathrm{SdAdd} * I)^2)$ is optimised, while indexing ambiguity resolution is performed using the underlying methods from *dials.cosym*.

4. Xia2.ssx: an automated processing pipeline

The individual DIALS programs described above form the building blocks of a processing pipeline for still-shot data. The pipeline for processing rotation data, as well as the analogous pipeline for still-shot data, is illustrated in Fig. 2. For still-shot processing, an accurate parameterisation of the detector is essential for reliable indexing and integration, whereas for

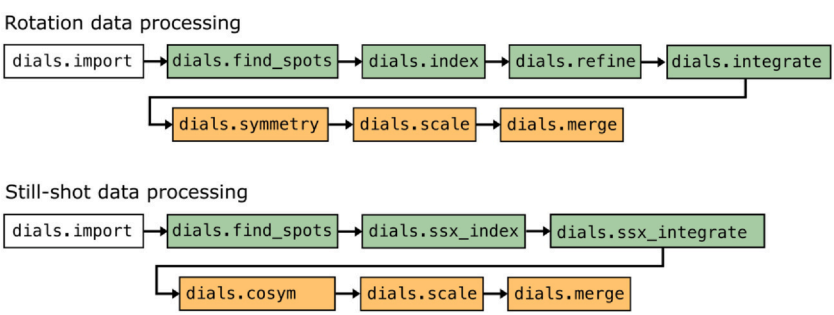

Fig. 2 The sequential processing pipeline for processing rotation and still-shot data with DIALS programs.

rotation data, this can be refined between indexing and integration. Non-isomorphism must be considered as part of the data reduction process, and consideration must be given to careful processing and separation of mixed-state serial data, such as time or dose series data. To handle these challenges, an automated processing pipeline was developed in the *xia2* suite, to provide a simple tool designed to handle these challenges and provide aggregated feedback to the user. For specialised analyses that do not fit into this model pipeline, the individual component programs in DIALS can be combined into alternative workflows.

4.1 Integration and reduction of SSX data with xia2.ssx

The *xia2.ssx* pipeline combines the individual DIALS programs into a workflow as illustrated in Fig. 3, which enables processing from raw images to merged data. The minimal required inputs for full processing are the image files, plus the expected unit cell and space group. The requirement for specifying a unit cell and space group is a practical consideration: in contrast to rotational crystallography, where the unit cell can reliably be determined through indexing the

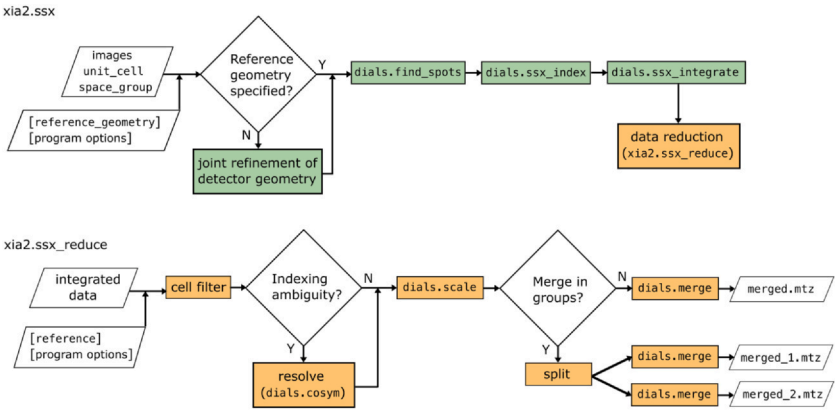

Fig. 3 A workflow diagram showing the processing steps of the *xia2.ssx* pipeline. The minimal required input are the image data, expected unit cell and space group. If no reference geometry is specified, a joint refinement of the detector geometry is performed, used indexed data from a subset of the images. Then, spot-finding, indexing and integration are performed in batches of 1000 images, before data reduction, which can be run as the standalone program *xia2.ssx_reduce*. In data reduction, a unit cell filter removes non-isomorphous crystals, before indexing ambiguity resolution using *dials.cosym*, if appropriate based on the crystal symmetry or cell parameters. Then each crystal is scaled, optionally in parallel batches against a reference set of intensities if specified. Finally, data can be split into groups based on experimental condition before each group is merged, producing a merged MTZ output file.

3D reciprocal lattice, and the space group symmetry can be discovered by testing the existence of rotational and screw axes, the existing algorithms are less reliable for still-image analysis due to the sparsity of reflections measured per crystal. In addition, serial crystallography experiments are typically performed to study particular aspects of well-characterised systems, so the unit cell, space group and reference intensity data are typically known at the start of data processing. If these parameters are not specified as input, the *xia2.ssx* pipeline will stop after spot-finding and indexing, outputting the most populous cell clusters and potential lattice symmetries for the user to consider and provide as input for a subsequent run.

Given this input, the first step is to determine an accurate model of the detector geometry. It has previously been shown that having a well-characterised experimental geometry improves the signal strength from serial crystallography data (Brewster et al., 2018), which can be determined from a joint refinement of the detector and crystal models, a feature that was already implemented within the *dials.refine* program and described in Brewster et al. (2018). The procedure used in *xia2.ssx* is to perform spot-finding (*dials.find_spots*) and indexing (*dials.ssx_index*) in batches of 1000 images, until a threshold number of crystals are indexed. Then a joint refinement of the crystal models and a single detector model are performed using *dials.refine* to determine the best-estimate of the detector geometry. This improved detector model is then fixed for indexing and integration of the full dataset. Alternatively, an external reference geometry can be given, either from previous DIALS processing of serial data or from DIALS processing of a rotation dataset. This can be used as a starting point for further joint refinement with the *starting_geometry* option, or as a fixed reference with no further joint refinement with the *reference_geometry* option. Following optimisation of the experimental geometry, spot-finding (*dials.find_spots*), indexing (*dials.ssx_index*) and integration (*dials.ssx_integrate*) are performed on batches of 1000 images. A summary of the processing statistics is logged at the completion of each process for each batch, with log files, HTML reports and output data from the individual programs output to batch subdirectories. Integration can be performed with the *ellipsoid* profile model (default) or *stills* integrator.

Following integration of all images, the first step of data reduction is to perform unit cell filtering, intended to remove a small fraction of the most non-isomorphous cells. For quick filtering, a filter based on the median unit cell values and absolute length (1.0 Å) and angle (1.0°) tolerances is applied by default. Alternatively, cells within a certain tolerance of a specified cell can be

selected, useful for selecting a subset of multi-modal cell distributions, or full clustering analysis can be triggered using *dials.cluster_unit_cell*, with the largest cluster being retained. Before scaling, an assessment is performed to determine whether indexing ambiguity resolution is required, by comparing the space group symmetry (used in indexing and integration) to the lattice symmetry. The lattice symmetry is determined from the unit cell values including a tolerance factor (the parameter *lattice_symmetry_max_delta* (Le Page, 1982)), which allows for the case of 'accidental' indexing ambiguities due to approximately equal cell parameters in addition to the 27 space groups where the symmetry of the Bravais lattice is higher than the space group symmetry. If indexing ambiguities are possible, then the *dials.cosym* program, run with the space group specified, is used to resolve the indexing ambiguities. The current implementation of the cosym algorithm has an n^2 memory dependence, where n is the number of crystals, therefore a modified procedure is used, where the indexing ambiguity is solved for batches of crystals at a time (default batch size 1000). Then, the crystals within each batch are scaled, before a final run of *dials.cosym* where each batch is considered as a single dataset, to consistently index all batches. If a reference file has been provided for scaling, the indexing solution chosen is that which best correlates with the reference. For scaling, when a reference file is provided, *dials.scale* is run on batches of crystals using the reference intensities in scaling (see Section 3.3), which places all crystals on a common scale across all batches. If no reference is provided, all crystals are scaled together in a single run of *dials.scale*, however this becomes computationally expensive for many thousands of crystals. Before the final merging step, the data can optionally be split into groups based on metadata, such as time-point or dose, as described below. Finally, *dials.merge* is used to merge the data for each group and produced a merged MTZ file. Throughout the data reduction pipeline, a *partiality_threshold* of 0.25 is applied.

The *xia2.ssx* program provides the most important program options as directly settable command line parameters, which are a limited subset of the full parameters available in the individual DIALS programs. To configure more advanced parameters for individual programs, *xia2.ssx* makes use of a feature of the cctbx ecosystem called 'phil' files; plain-text files that specify program parameter values, which can be specified for the individual DIALS programs (an example is given in Section 5.2). As such, the *xia2.ssx* program defines a processing workflow and acts as a runner for the individual DIALS programs, setting suitable default program parameters for SSX datasets while allowing full user specification of program parameters.

4.2 Standalone data reduction with xia2.ssx_reduce

To enable quick reduction and aggregation of integrated data, the data reduction part of the *xia2.ssx* pipeline can be run as the standalone command *xia2.ssx_reduce*, which takes DIALS-integrated experiment and reflection files as inputs. If several datasets have been reduced separately but with the same reference set of intensities, then all crystals will be on the same scale and the scaled data can quickly be merged without repeating data reduction. To achieve this, the *xia2.ssx_reduce* program can be run with the option *steps=merge*. Fig. 4 shows an example workflow for processing and aggregating blocks of images in an efficient manner. With a reference geometry predetermined from a well-characterized sample, images can be integrated with a consistent detector geometry and scaled against a reference set of intensities. The output DIALS scaled files can then be used as input to *xia2.ssx_reduce*, with the option *steps=merge* triggering solely the merging step, as the scaled data are already on a common scale fixed by the reference. If no reference intensities were available for processing, the alternative workflow would be to use the DIALS integrated files as input to *xia2.ssx_reduce*, which would perform the full data reduction process.

4.3 Merging in groups based on metadata

Serial crystallography enables a new kind of experiment, where repeated measurements are taken on a given crystal, to measure the evolution with time after X-ray exposure and/or laser-excitation, giving a dose or time series with a defined repeat interval. For cases where the resultant change in diffraction intensities is expected to be modest, one workflow is to process all data together, to get data processed in a consistent manner and placed on a common scale, before separately merging based on experimental

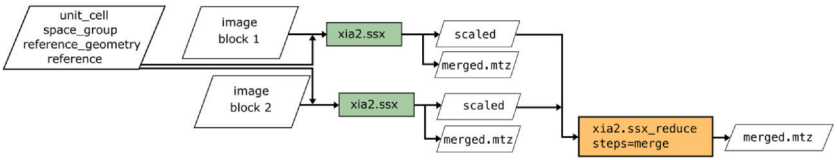

Fig. 4 Schematic of typical workflow for processing experimental data, allowing quick merging of sets of images after initial processing using a reference set of intensities. The final processing step (*xia2.ssx_reduce*) only needs to merge the scaled and consistently indexed data.

condition. *xia2.ssx* contains a mechanism to allow splitting of scaled data before merging in separate groups. In *xia2.ssx/ xia2.ssx_reduce*, the option *dose_series_repeat=n* can be used to define the repeat interval *n* that should be used to split the data, i.e. every *n*th image is part of the same experimental condition. For generalised merging of images into specified groups in an automated manner based on experimental metadata, a YAML configuration file (as illustrated in Listing 1 below) can be used to define the location of metadata for grouping images for merging. For example, consider a HDF5 image file containing a mixture of pump and probe diffraction data, pre-filtered for empty images, where the pump status (on/off) is written into the file as a data array.

When supplied with a correctly specified YAML file, *xia2.ssx/ xia2.ssx_reduce* will split the scaled data into groups based on the pump status label and merge each group separately. Importantly, as the splitting is performed by inspecting the image index via the experiment list data structure, it does not rely on the scaled data being ordered and can therefore be used when merging multiple datasets together, such as in the workflow shown in Fig. 4.

Listing 1: Example definition of a *xia2.ssx* grouping YAML file, to trigger merging in groups based on the 'pump status' label.

```
---
  metadata:
    pump:
      path/to/image1.h5: path/to/image1.h5:/
      entry/pump_status
      path/to/image2.h5: path/to/image2.h5:/
      entry/pump_status
  grouping:
    merge_by:
      values:
        - pump
---
```

This provides a mechanism by which data acquisition software can write metadata into an image or metadata file, which can automatically be used by *xia2.ssx* to produce grouped output data. Alternatively, post-processing can be used to determine a classifier, which when stored as a HDF5 array can be used by *xia2.ssx* to divide images into groups for merging.

5. Demonstration and evaluation on example data

5.1 Processing anomalous SSX data: AcNiR above the K-edge

A dataset consisting of 19,200 images was collected on a fixed-target chip at beamline I24, Diamond Light Source, containing microcrystals of copper nitrite reductase from *Achromobacter cycloclastes* (AcNiR). The collection wavelength was 1.376 Å, just above the Cu K-edge, and therefore a high level of anomalous signal was expected. AcNiR is known to crystallise into the $P2_13$ cubic space group with a cell length 96.4 Å, with a polymorph with $a = b = c = 97.8$ Å (Ebrahim et al., 2019).

In this demonstration, we first process the data with the *xia2.ssx* pipeline and discuss the processing steps. We then evaluate the ellipsoid profile model by comparison with the stills profile model, by reducing integrated data with the *cctbx.xfel.merge* program, so that post-refinement can be applied to the stills profile model as is standard practice. Structure refinement is performed with *REFMAC5* (Murshudov et al., 2011) via *DIMPLE* v2.6.2 (https://ccp4.github.io/dimple/). A reference model from PDB entry 2BW4 (Antonyuk et al., 2005) was used, enabling a consistent free set of reflections, and the *–anode* option was used to run *ANODE* (Thorn & Sheldrick, 2011) for calculating anomalous difference maps and anomalous peak heights.

5.1.1 Standard processing with xia2.ssx

The raw data consists of 19,200 images in CBF format collected on a Pilatus 6 M, named sequentially from image_00000.cbf to image_19199.cbf. The full command to process the data with *xia2.ssx* was.

```
xia2.ssx template=images/image_#####.cbf \.
  mask=pixels.mask \.
  geometry_refinement.n_crystals=2000 \.
  space_group=P213 \.
  unit_cell=96.4,96.4,96.4,90,90,90 \.
  anomalous=True \.
  d_min=1.8 \.
  reference=2bw4.pdb \.
  absolute_length_tolerance=0.3.
```

A mask was generated manually in the *dials.image_viewer* to mask the beam-stop shadow. A minimum of 2000 indexed crystals were required for

the initial geometry refinement, and a reference model from PDB entry 2BW4 was used to generate a reference intensity set for data reduction. During the initial geometry refinement routine, spot-finding and indexing were performed on the first four batches of 1000 images, at which point 2215 crystals had been indexed, with a median cell constant of 96.66 ± 0.14 Å. During the geometry refinement, the spots RMSDs reduced from (0.309 mm, 0.163 mm, 0.126 °) to (0.141 mm, 0.085 mm, 0.061 °). Subsequently, the processing of the first four batches with the refined geometry gave 2403 indexed crystals, with a median cell constant of 96.39 ± 0.15 Å. During indexing and integration, 56 % of the images were indexed with at least one lattice, with multiple lattices per image being common, resulting in 16,440 integrated crystals. The average sigmas of the ellipsoid mosaicity model were (161, 378, 497) $\mu Å^{-1}$, i.e. an average ellipsoidal aspect ratio of 3.1, indicative of significant diffraction spot anisotropy as could be seen on the raw images.

A tighter unit cell filter tolerance was applied (*absolute_length_tolerance=0.3*) for the purposes of data reduction, to avoid inclusion of the polymorph cell, which filtered out 274 crystals. As the $P2_13$ space group point group symmetry is lower than the $P432$ lattice symmetry, the indexing ambiguity must be resolved by the resolve aspect of the pipeline, the details of which were discussed in Section 4. Fig. 5A shows the *dials.cosym* output from a representative batch of 1000 crystals, demonstrating clear separation of the two indexing modes for most crystals. The output from reindexing all batches with *dials.cosym* is shown in Fig. 5B, again showing clear clustering indicating resolution of the indexing modes. Another means to confirm the success of indexing ambiguity resolution is the Padilla-Yeates L-test (Padilla & Yeates, 2003), which is performed as part of the merging procedure in *dials.merge* and shown in Fig. 5C. In this case the L-statistic more closely matches the theoretical untwinned curve than the perfect twin curve, suggesting the data is not perfectly untwinned, consistent with the small fraction of crystals lying between the two main cosym clusters in Fig. 5A.

The overall merging statistics showed high internal consistency, with $CC_{1/2} = 0.997$, $R_{split} = 0.055$, $I/\sigma = 20.6$, with an anomalous multiplicity of 125.0. The final R/R_{free} values were 0.112/0.147 and the Cu anomalous peak heights were 50.3 σ & 44.2 σ. As a comparison, the data were also reduced without the use of a reference, which resulted in lower internal consistency, with $CC_{1/2} = 0.982$, $R_{split} = 0.083$, $I/\sigma = 17.1$, with an anomalous multiplicity of 137.9. The final R/R_{free} values were 0.165/0.210

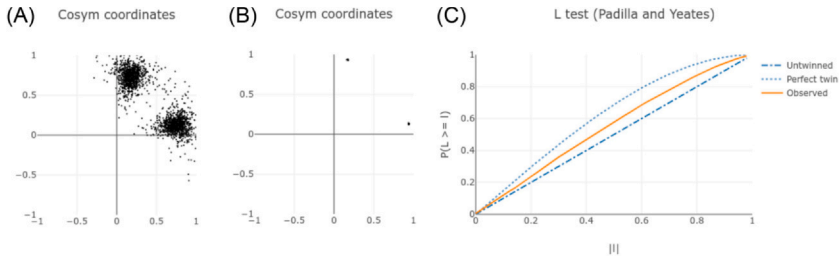

Fig. 5 Indexing ambiguity resolution of SSX data with *dials.cosym*. (A) output cosym coordinates plot for a representative batch of 1000 crystals, showing separation into two main clusters for most crystals. (B) indexing ambiguity resolution of all 16 batches, which separate into two tight clusters. (C) L-test metric plot for twinning; for this dataset the statistic does not exactly match the theoretical untwinned curve, suggesting that the data is not perfectly untwinned.

and the Cu anomalous peak heights were 34.6 σ & 29.1 σ. Aside from the reduced runtime, the main benefit of the reference model is to aid in outlier rejection, with the highest fraction of outliers occurring at lower resolutions, leading to stronger anomalous signal and significantly improved model R-factors.

5.1.2 Comparison with existing methods

To evaluate the ellipsoid partiality model, the data were processed using both the *ellipsoid* and *stills* integration algorithm. In addition to the different partiality estimates of the two algorithms, the set of reflections predicted also differs. To test the differences in partiality estimates, we first performed a comparison using a common set of reflections predicted by both integration algorithms. As the *stills* integration algorithm has been developed for use with post-refinement methods available in *cctbx.xfel.merge*, the base comparison case is to perform data reduction with *cctbx.xfel.merge*, enabling post-refinement for stills-integrated data and disabling post-refinement for ellipsoid-integrated data. Next, the differences in prediction were assessed by processing the full ellipsoid-integrated dataset with *cctbx.xfel.merge*. Finally, we compared processing of the full ellipsoid-integrated dataset reduced with *xia2.ssx_reduce* with the full stills-integrated dataset reduced with *cctbx.xfel.merge*.

A common set of stills and ellipsoid integrated reflections were chosen by matching the miller indices for each crystal in each image, for those reflections with an ellipsoid-integrated partiality estimate $p > 0.25$. For data reduction with *cctbx.xfel.merge*, the ellipsoid-integrated intensities were

corrected for partiality and the partiality uncertainty was propagated to the intensity uncertainties before the data reduction. A reference model from PDB entry 2BW4 was used, anomalous pairs were separated and a resolution limit of 1.8 Å was applied. Fig. 6 shows the *REFMAC R*-factors and *ANODE* anomalous peaks heights for the data processed cumulatively in batches of 1000 crystals, up to a total of 16,166 crystals. The ellipsoid data R_{free} values are consistently lower than the stills-integrated data by around one percentage point. Alternatively, a given R/R_{free} is reached with many fewer crystals for the ellipsoid-integrated data. The Cu anomalous peak heights also show consistent improvement for the ellipsoid integrated data, indicating a more accurate anomalous signal. Processing statistics for the full 16,166 crystals are shown in Table 1. For this dataset, the ellipsoid model gives a clearly improved partiality estimate compared to post-refining a spherical model. We attribute this to the fact that the spot shapes have significant anisotropy, therefore there is limited improvement that can be gained from post-refining an isotropic mosaicity model.

Next, the full ellipsoid-integrated dataset ($p > 0.25$) was reduced with *cctbx.xfel.merge*, to investigate the effect of the additional predictions of the ellipsoid model, which increased the anomalous multiplicity from 98.8 to 141.4 for the full dataset (16,166 crystals). Overall, as listed in Table 1, the full ellipsoid-integrated dataset processed with *cctbx.xfel.merge* had similar data quality to the common subset dataset. The overall I/σ increased slightly from 8.65 to 8.82, with R_{split} lowered from 5.8 % to 5.5 %. R_{free} increased slightly

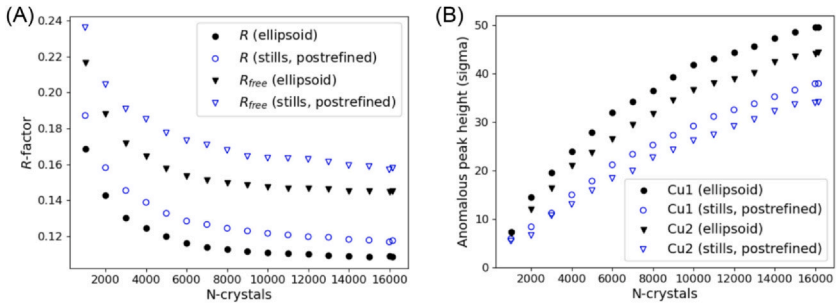

Fig. 6 A comparison of processing the AcNiR SSX dataset with the two integration algorithms, ellipsoid and stills, for a common set of reflections reduced with *cctbx.xfel.merge*. (A) *REFMAC* R/R_{free} and (B) *ANODE* Cu anomalous peak heights, for the dataset processed cumulatively in batches of 1000 crystals. For this dataset, the ellipsoid partiality model results in consistently improved *R*-factors and anomalous peak heights, which therefore enables a structure of a particular quality to be achieved with significantly fewer measured crystals.

Table 1 Comparative statistics for processing the AcNiR dataset with the stills and ellipsoid integration algorithms, reduced with cctbx.xfel.merge.

Profile/integration algorithm	**Stills**	**Ellipsoid**	**Ellipsoid**
Reduction program	cctbx.xfel.merge	cctbx.xfel.merge	cctbx.xfel.merge
Post-refinement enabled	Yes	No	No
Common subset	Yes	Yes	No
$CC_{1/2}$	0.997	0.996	0.996
*R*split	0.066	0.058	0.055
I/σ	6.60	8.65	8.82
Anom. multiplicity	60.5	98.8	141.4
R/R_{free}	0.117/0.158	0.109/0.145	0.111/0.147
wR/wR_{free}	0.122/0.167	0.116/0.157	0.119/0.158
Anom. peak heights	38.0, 34.1	49.6, 44.4	52.1, 45.5
Error model (SdFac, SdB, SdAdd)	1.31, 0.33, 0.40	1.01, 0.10, 0.64	0.97, 0.14, 0.79

The first two columns are statistics for a common subset of reflections integrated by both algorithms; the final column is the full ellipsoid-integrated dataset. This shows that the improved data quality for the ellipsoid model comes primarily from improved partiality estimates, rather than additional predicted reflections.

from 0.145 to 0.147 and the peak heights increased slightly from (49.6, 44.4) to (52.1, 45.5). The trends for the data processed cumulatively (not shown) showed similar behaviour, with R_{free} and anomalous heights marginally higher. Looking at the overall trends, it is clear that the improvement in data quality is coming primarily from the improved partiality estimates rather than the additional predicted reflections.

The statistics for the full ellipsoid-integrated dataset (16,166 crystals, $p > 0.25$), reduced with *xia2.ssx_reduce* are shown in Table 2, alongside statistics for the full stills-integrated dataset reduced with *cctbx.xfel.merge*. We cannot directly compare with the ellipsoid-integrated dataset reduced with *cctbx.xfel.merge*, as the disabling of post-refinement means that outliers cannot be removed during data reduction. In *xia2.ssx_reduce*, outlier rejection is performed in the scaling step, which reduces the anomalous multiplicity from 141.4 to 125.0, whereas in *cctbx.xfel.merge*, reflections can only be removed as

Table 2 Comparative statistics for processing the AcNiR dataset for the full stills-integrated dataset reduced with cctbx.xfel.merge and the full ellipsoid-integrated dataset reduced with xia2.ssx_reduce.

Profile/integration algorithm	Stills	Ellipsoid
Reduction program	cctbx.xfel.merge	xia2.ssx_reduce
Post-refinement enabled	Yes	No
$CC_{1/2}$	0.996	0.997
*R*split	0.068	0.055
I/σ	6.5	20.6
Anom. multiplicity	63.3	125.0
R/R_{free}	0.117/0.160	0.112/0.147
wR/wR_{free}	0.121/0.168	0.100/0.133
Anom. peak heights	37.4, 33.9	50.3, 44.2
Error model (SdFac, SdB, SdAdd)	1.36, 0.31, 0.39	1.17, N.A., 0.08[a]

[a]average over data reduction batches.

part of re-prediction in post-refinement. The ellipsoid-integrated dataset reduced with *xia2.ssx_reduce* results in lower *REFMAC R*-factors and higher anomalous peak heights compared to the stills-integrated dataset reduced with *cctbx.xfel.merge*, consistent with the previous comparison on the common subset of reflections. Another notable difference is the difference in I/σ values, primarily due to the different error model corrections optimised during data reduction. In *xia2.ssx_reduce*, the *dials.scale* error model parameters refined to much lower values, i.e. the σ values are not increased to the same extent, giving higher relative I/σ. The ellipsoid-integrated σ values are additionally increased by the partiality uncertainty.

To verify the reliability of the σ estimates from *xia2.ssx_reduce*, it is instructive to look at the σ-weighted *R*-factors (wR/wR_{free}) from *REFMAC* (see Table 2). For the stills-integrated data post refined with *cctbx.xfel.merge*, $wR/wR_{free} = 0.121/0.168$, whereas for the ellipsoid-integrated reduced data with *xia2.ssx_reduce*, $wR/wR_{free} = 0.100/0.133$. The differences between the weighted *R*-factors are much larger than for the unweighted *R*-factors, indicating more reliable σ estimates from the new methods. Fig. 7 shows wR/wR_{free} by resolution for the stills-integrated

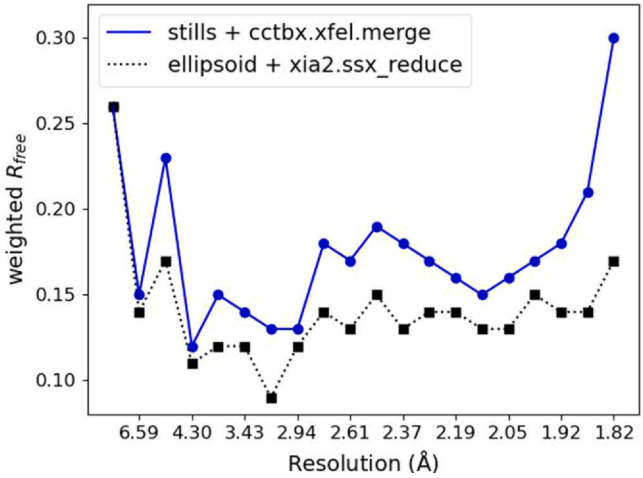

Fig. 7 *REFMAC σ*-weighted R_{free} by resolution for AcNiR data processing, for the full stills-integrated dataset reduced with *cctbx.xfel.merge* (blue, circles) and the full ellipsoid-integrated dataset reduced with *xia2.ssx_reduce* (black, squares). The *xia2.ssx_reduce* processing of the ellipsoid-integrated data gives significantly lower weighted R_{free} values, except in the lowest resolution shells. Both methods give a high R_{free} at the lowest resolution shell, indicating a general deficiency in modelling the lowest resolution reflections.

data, reduced with *cctbx.xfel.merge*, and the ellipsoid integrated data reduced with *xia2.ssx_reduce*. The wR_{free} values for processing with *xia2.ssx_reduce* are significantly lower, except in the lowest resolution shells. The high *R*-factors at the lowest resolution indicate a general deficiency in modelling the lowest resolution reflections and should be a key focus of further algorithm development.

5.2 Processing SSX dose-series: photoreduction of Fe(III) FutA

A room-temperature SSX dose-series dataset consisting of 256,000 diffraction images of the FutA iron-binding protein was analysed. This was collected as 10 measurements of repeated exposure equivalent to 22 kGy dose at each location on a fixed-target chip at beamline I24, Diamond Light Source, and was previously published in Bolton et al. (2024). This previous study confirmed the presence of an Fe3+ state, with subsequent photo-reduction of the Fe3+ site accompanied by a change in conformation of the Arg203 residue, which was observed to be visible at 22 kGy and strongest at 88 kGy. In this work, we demonstrate the features available in

xia2.ssx/DIALS to process images collected under different experimental conditions. Although this example is a dose-series dataset, the methods are applicable in general to other types of serial diffraction measurements investigating dynamics such as laser-excitation series, X-ray pump-probe or ligand-binding studies.

The raw data consists of 256,000 images in CBF format, collected on a Pilatus 6 M, were sequentially named from image_000000.cbf to image_255999.cbf. First, a reference geometry was refined with the command:

```
xia2.ssx template=images/image_######.cbf \.
  dials_import.phil=import.phil \.
  mask=pixels.mask \.
  space_group=P21 \.
  unit_cell=39.7,78.7,48.4,90,97.8,90 \.
  geometry_refinement.n_crystals=2000 \.
  steps=None.
```

In the raw image headers, the beam centre was incorrectly specified by around 30 pixels along the fast axis, therefore this was overwritten on import by providing a 'phil' file for the *dials.import* program, containing the line *geometry.detector.fast_slow_beam_centre=1292,1310*. A mask was generated manually in the *dials.image_viewer* to mask the beam-stop shadow, while *steps=None* instructs no full processing to be done after geometry refinement. Then, due to a significant tilt in the detector orientation, another round of geometry refinement was performed at the start of full processing, starting with the geometry refined from the first process (contained in the *refined.expt* file), with the command:

```
xia2.ssx template=images/image_######.cbf \.
  mask=pixels.mask \.
  space_group=P21 \.
   unit_cell=39.7,78.7,48.4,90,97.8,90 \.
  starting_geometry=geometry_refinement/refined.expt \.
  geometry_refinement.n_crystals=2000 \.
  steps=find_spots+index+integrate.
```

The change in detector geometry, from initial to final refined geometry, is shown in Fig. 8A, which shows a relative displacement and tilt. During indexing and integration, 25.1 % of images were indexed with at least one lattice. In total, 96,612 lattices were independently integrated. The repeated measurements are currently treated independently in indexing and

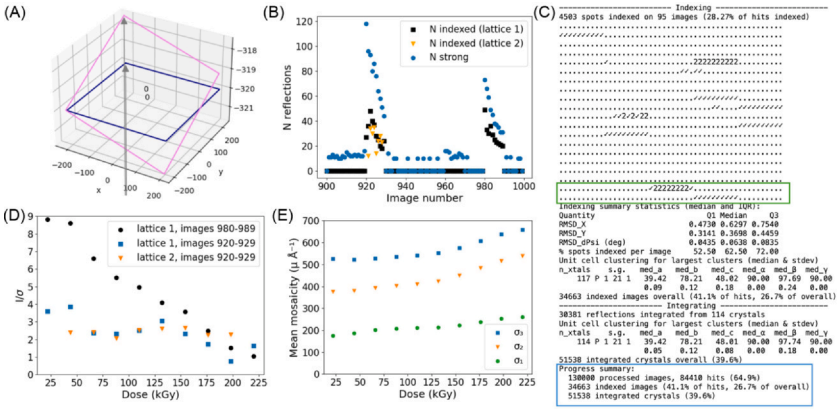

Fig. 8 Sample output for FutA SSX dose series data processing. (A) Comparison of detector models in the image headers (dark blue) and after joint geometry refinement (pink), showing a relative displacement and tilt. (B) Example indexing plot from a *dials.ssx_index.html* report, showing trends in the number of strong spots and indexing success for diffraction at images 920–929 and 980–989 as 10 measurements are made at each location on the chip. (C) A subsection of the *xia2.ssx* log output for indexing and integration of this batch of 1000 images, with the 100-image range from (B) highlighted by the added green box. Over the 1000 images, some physical crystals are successfully indexed on each image, but this is not always the case: '.' indicates an unindexed image, a tick is a singe lattice indexed and '2' is two lattices indexed on an image. A cumulative summary is also presented after each batch (added blue box). (D) Mean I/σ by dose, based on data in the *dials.ssx_integrate.html* report, for the indexed crystals from (B). A clear reduction in *I/σ* is seen for the crystal in images 980–989, whereas lattice 2 in images 920–929 has no evident change in I/σ with dose. (E) The average ellipsoid mosaicity components per dose point, calculated over the whole dataset after data reduction. The mosaicity increases gradually from 22 to 110 kGy, after which the increase is more pronounced.

integration resulting in up to 10 integrated lattices for each physical crystal. A selection of example output is show in Fig. 8B–D.

To scale all integrated data together before splitting and merging each dose point, the *xia2.ssx_reduce* command was used:

```
xia2.ssx_reduce batch_*/integrated* d_min=2.1 dose_series_repeat=10.
```

In this case, we took advantage of the fact that there is a repeating pattern of experimental conditions by using the option *dose_series_repeat=10* to instruct merging into 10 groups based on the image index. A *d_min* value of 2.1 Å was used for data reduction to be consistent with the previously published data. No reference was used for data reduction in this example. 96,080 lattices remained

after data reduction, which gave between 11,981 (66 kGy) and 6297 (220 kGy) lattices per dose point. For this dataset, there is no indexing ambiguity due to symmetry or accidental ambiguity due to similar lattice parameters. The merged MTZ file for each dose point was used as input for structure determination with *REFMAC5* via *DIMPLE*, with a starting model of PDB entry 8OGG (Bolton et al., 2024).

Table 3 shows a selection of data quality metrics for this processing, where all dose points are scaled together. Above 110 kGy, there is a reduction in I/σ that is indicative of radiation damage and cannot be accounted for solely by the reduced number of integrated crystals or reduction in multiplicity. In addition, the relative Wilson *B*-factor increases steadily up until the 176 kGy dose point. As a relative *B*-factor is applied to each image in scaling, data with a small amount of global radiation damage can in theory be corrected, such that the Wilson *B*-factors, when calculated from data placed on the same relative scale, would be approximately constant. Therefore, the increase in relative *B*-factor indicates that even at low doses the effect of radiation damage is causing a reduction of intensity with resolution to a greater extent than that which can be corrected for by a simple global *B*-factor correction. The effect of radiation damage can also be seen in the average mosaicity model parameters (Fig. 8E), which increase as a function of dose, particularly towards the end of the dose series.

To investigate the lower dose points, the data from 22 kGy to 88 kGy were scaled together. To achieve this, first the integrated data were split into separate files by dose using the *dials.split_still_data* command, then *xia2.ssx_reduce* was run on files only containing the first four doses.

```
dials.split_still_data batch_*/integrated*
series_repeat=10.
xia2.ssx_reduce group_{0,1,2,3}* d_min=2.1
dose_series_repeat=10.
```

Table 4 shows the processing statistics for this processing. The merging statistics for the four dose points are overall similar to when all ten dose points were processed together, with I/σ and multiplicity increased due to fewer reflections being rejected as outliers. Fig. 9 shows $2F_o - F_c$ electron density maps for the 22 kGy and 88 kGy datasets, as well as $F_o - F_c$ difference density maps. The density maps indicate a partial conformational change of the Arg203 residue, which is more pronounced at 88 kGy, as shown in the previous study (Bolton et al., 2024). Furthermore, at 88 kGy there is loss of density at the water site above the Fe and loss of density at the Tyr13 residue, i.e. site-specific changes have occurred.

Table 3 Summary merging statistics for ten-point FutA SSX dose series processed with xia2.ssx_reduce.

Dose (kGy)	22	44	66	88	110	132	154	176	198	220
n-crystals	10,149	11,395	11,981	11,623	10,960	9832	8170	7980	7153	6297
Multiplicity	34.9	41.6	46.9	47.2	45.2	41.2	38.1	37.2	35.4	32.2
I/sigma	12.3	13.7	14.0	13.1	11.7	10.1	8.5	7.3	6.2	5.3
$CC_{1/2}$	0.89	0.91	0.92	0.92	0.92	0.93	0.93	0.92	0.93	0.93
R_{split}	0.19	0.17	0.16	0.16	0.16	0.17	0.17	0.17	0.18	0.18
Wilson B (Å^2)	11.7	13.8	16.4	19.2	22.0	24.7	26.8	27.8	28.1	28.0
R_{work}	0.192	0.193	0.193	0.194	0.192	0.189	0.185	0.180	0.175	0.173
R_{free}	0.231	0.229	0.232	0.234	0.238	0.231	0.228	0.220	0.222	0.218

The resolution range for all datasets is 47.6 Å to 2.1 Å.

Table 4 Summary statistics for combined reduction of the first four dose points of the FutA SSX dose series processed with xia2.ssx reduce (resolution range 47.6 Å to 2.1 Å).

Dose (kGy)	22	44	66	88
n-crystals	10,149	11,396	11,981	11,621
Multiplicity	36.7	43.2	48.2	48.2
I/sigma	13.3	14.6	14.7	13.6
$CC_{1/2}$	0.90	0.92	0.93	0.93
R_{split}	0.19	0.17	0.16	0.16
Wilson B (Å^2)	11.4	13.7	16.6	19.6
R_{work}	0.197	0.197	0.196	0.195
R_{free}	0.236	0.233	0.234	0.233

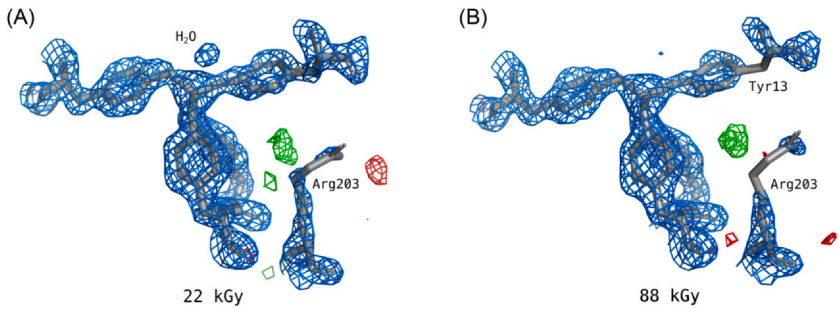

Fig. 9 Electron density maps of the Fe environment ($2F_o - F_c$, blue, 1.5σ) and difference density ($F_o - F_c$, green, 3σ) for FutA after accumulation of (A) 22 kGy and (B) 88 kGy. In both structures, the difference densities indicate partial conformational change of the Arg203 residue, which is greater at 88 kGy, consistent with results in Bolton et al. (2024). At 22 kGy, density is still present at the water above the Fe site, which is absent at 88 kGy, while loss of density at the Tyr13 residue is evident at 88 kGy dose. Images were generated with PyMOL (DeLano et al., 2002).

In this section, we demonstrated how mixed-state data can be scaled together with *xia2.ssx* before merging, and the tools that are available to split mixed-state data before scaling and merging. The purpose was to demonstrate the available tools rather than provide a recommendation for one correct approach to handle all possible experimental cases. The correct approach will depend on the level of isomorphism between the different experimental conditions and the scientific question being addressed by the

experiment. Further algorithm developments could improve processing for repeated measurements on the same crystals, such as performing joint indexing on serial images from the same physical crystal to improve indexing success across the series of images. In the cases of indexing ambiguity resolution, this data could then be considered a single entity for reindexing.

6. Discussion

In this work, we have demonstrated the methods available in DIALS and *xia2* for processing serial still-shot diffraction data. This includes a generalised anisotropic model to describe the distribution of diffraction intensity about reciprocal lattice points, an empirical approach which can describe anisotropic mosaic block sizes and orientational distributions. This e*llipsoid* model is optimised using a maximum-likelihood algorithm that simultaneously considers the pixel intensity distribution, reflection offsets, and spot shapes. In addition to providing more accurate partiality estimates, this approach also offers the advantage of estimating the standard error of the partiality. This allows for more reliable weighting of integrated intensities during data reduction and merging. Currently only a monochromatic beam model has been implemented in DIALS, which is suitable for SSX but will have lower fidelity for pink-beam SSX or SFX. The inclusion of wavelength spread as part of the profile model optimisation and integration (Parkhurst, 2020), could bring further data quality improvements for those experiments.

We described and demonstrated some of the workflows available in the *xia2.ssx* and *xia2.ssx_reduce* programs for processing serial still-shot diffraction data. This includes automated detector geometry refinement, workflows for handling scaling and merging of data as well as the means to manage mixed-state datasets such as dose or time series datasets. As the right processing approach will depend on the dataset and experiment in question, the aim was to demonstrate the features available and the use of the program outputs to evaluate the data processing steps, which we encourage any users of the programs to become familiar with. We demonstrated an example dataset where the ellipsoid integration algorithm gives improvements upon isotropic models, however there can still be a significant fraction of outliers after integration. In these cases, the use of a reference intensity set during scaling is beneficial for outlier rejection. For typical

SSX experiments, where the purpose of the study is to investigate dynamics of a known structure, suitable reference intensity sets are typically available. Currently we do not include a means to post-refine the ellipsoid profile model, which could further improve the final data quality. To post-refine the ellipsoid model would require rewriting the scaling objective function in a likelihood formalism, which could then be maximised in combination with the likelihood function of the ellipsoid model (Eq. 9). This contrasts with post-refinement methods which minimise the deviation between predicted and observed spot positions. However, post-refinement of the ellipsoid model would have a much higher requirement on computational resources than the integration step, as it would require log-likelihood optimisation not limited to only the strong reflections, and must be done for all crystals together, compared to integration which is done individually per crystal. Another important aspect of SSX processing is the handling of non-isomorphism. *xia2.ssx* contains methods to handle cell non-isomorphism and indexing non-isomorphism, while DIALS includes further tools to investigate intensity non-isomorphism, such as $\Delta CC_{1/2}$ analysis (Winter et al., 2022). The ability to use the output of the *xia2.ssx* pipeline for further analysis provides flexibility in the analysis that can be performed with DIALS and *xia2*, enabling the user to tailor the analysis to tackle technique-specific or dataset-specific challenges in serial synchrotron crystallography.

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Appendix A. Block matrix inversion

If a matrix, A, can be partitioned as:

$$\boldsymbol{A} = \begin{pmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{pmatrix} \tag{15}$$

then, as shown in Petersen & Pedersen (2012), its inverse is

$$\boldsymbol{A} = \begin{pmatrix} \boldsymbol{A}^{11} & \boldsymbol{A}^{12} \\ \boldsymbol{A}^{21} & \boldsymbol{A}^{22} \end{pmatrix} \tag{16}$$

where

$$\begin{aligned} \boldsymbol{A}^{11} &= (\boldsymbol{A}_{11} - \boldsymbol{A}_{12}\boldsymbol{A}_{22}^{-1}\boldsymbol{A}_{21})^{-1} = \boldsymbol{A}_{11}^{-1} + \boldsymbol{A}_{11}^{-1}\boldsymbol{A}_{12}\,(\boldsymbol{A}_{22} - \boldsymbol{A}_{21}\boldsymbol{A}_{11}^{-1}\boldsymbol{A}_{12})^{-1}\boldsymbol{A}_{21}\boldsymbol{A}_{11}^{-1} \\ \boldsymbol{A}^{12} &= -(\boldsymbol{A}_{11} - \boldsymbol{A}_{12}\boldsymbol{A}_{22}^{-1}\boldsymbol{A}_{21})^{-1}\boldsymbol{A}_{12}\boldsymbol{A}_{22}^{-1} = (\boldsymbol{A}^{21})^T \\ \boldsymbol{A}^{21} &= -\boldsymbol{A}_{22}^{-1}\boldsymbol{A}_{21}\,(\boldsymbol{A}_{11} - \boldsymbol{A}_{12}\boldsymbol{A}_{22}^{-1}\boldsymbol{A}_{21})^{-1} = (\boldsymbol{A}^{12})^T \\ \boldsymbol{A}^{22} &= \boldsymbol{A}_{22}^{-1} + \boldsymbol{A}_{22}^{-1}\boldsymbol{A}_{21}\,(\boldsymbol{A}_{11} - \boldsymbol{A}_{12}\boldsymbol{A}_{22}^{-1}\boldsymbol{A}_{21})^{-1}\boldsymbol{A}_{12}\boldsymbol{A}_{22}^{-1} = (\boldsymbol{A}_{22} - \boldsymbol{A}_{21}\boldsymbol{A}_{11}^{-1}\boldsymbol{A}_{12})^{-1} \end{aligned} \tag{17}$$

Appendix B. Derivatives of the log-likelihood and Fisher Information matrix

The Fisher scoring algorithm uses the Fisher Information matrix **I**, which is the negative of the expected value of the Hessian Matrix. The Fisher Information matrix is guaranteed to be positive semi-definite, leading to more robust convergence. Additionally, only the first derivatives need to be calculated, resulting in a less computationally intensive implementation (Osborne, 1992). In the Fisher method, at each iteration, $(t+1)$, the parameters $\boldsymbol{\beta}$ are updated according to

$$\boldsymbol{\beta}\,(t+1) = \boldsymbol{\beta}\,(t) + I\,(\boldsymbol{\beta}\,(t))^{-1}\,V\,(\boldsymbol{\beta}\,(t)), \tag{18}$$

where $V(\boldsymbol{\beta}_{(k)})$ is known as the score function and the elements of the Fisher Information matrix **I** are given by

$$I_{kl} = -E\left[\frac{\partial^2 L}{\partial\beta_k\,\partial\beta_l}\right] \tag{19}$$

where L is the log-likelihood.

As shown in (Petersen & Pedersen, 2012), the derivative of the determinant, trace and inverse of a matrix, A, are given by:

$$\frac{\partial}{\partial x}\,|A| = |A|\,\mathrm{tr}\left(A^{-1}\frac{\partial A}{\partial x}\right),\ \frac{\partial}{\partial x}\mathrm{tr}(A) = \mathrm{tr}\left(\frac{\partial A}{\partial x}\right),\ \frac{\partial}{\partial x}A^{-1} = -A^{-1}\frac{\partial A}{\partial x}A^{-1}. \tag{20}$$

Using these identities, the first derivatives of the log-likelihood function in Eq. 11 with respect to a parameter β_k are given by

$$\begin{aligned} \frac{\partial L}{\partial\beta_k} = &-\frac{1}{2}\sum_i^N c_{\mathrm{tot},i}\left[\Sigma_{22i}^{-1}\frac{\partial\Sigma_{22i}}{\partial\beta_k} - \Sigma_{22i}^{-2}\frac{\partial\Sigma_{22i}}{\partial\beta_k}\epsilon_i^2 + 2\Sigma_{22i}^{-1}\epsilon_i\frac{\partial\epsilon_i}{\partial\beta_k}\right] \\ &-\frac{1}{2}\sum_i^N c_{\mathrm{tot},i}\left[\mathrm{tr}\left(\bar{\boldsymbol{\Sigma}}_i^{-1}\frac{\partial\bar{\boldsymbol{\Sigma}}_i}{\partial\beta_k} - \bar{\boldsymbol{\Sigma}}_i^{-1}\frac{\partial\bar{\boldsymbol{\Sigma}}_i}{\partial\beta_k}\bar{\boldsymbol{\Sigma}}_i^{-1}\left(\mathbf{S}_i + (\bar{\mathbf{x}}_i - \bar{\boldsymbol{\mu}}_i)(\bar{\mathbf{x}}_i - \bar{\boldsymbol{\mu}}_i)^T\right)\right)\right] \\ &-\frac{1}{2}\sum_i^N c_{\mathrm{tot},i}\left[2\mathrm{tr}\left(\bar{\boldsymbol{\Sigma}}_i^{-1}\left((\bar{\mathbf{x}}_i - \bar{\boldsymbol{\mu}}_i)\frac{\partial\bar{\boldsymbol{\mu}}_i^T}{\partial\beta_k}\right)\right)\right] \end{aligned} \tag{21}$$

The derivatives of $\bar{\boldsymbol{\Sigma}}$, $\bar{\boldsymbol{\mu}}$, Σ_{22} and ϵ are given by

$$\begin{aligned}
\frac{\partial \bar{\boldsymbol{\Sigma}}}{\partial \beta_k} &= \frac{\partial \boldsymbol{\Sigma}_{11}}{\partial \beta_k} - \frac{\partial \boldsymbol{\Sigma}_{12}}{\partial \beta_k}\Sigma_{22}^{-1}\boldsymbol{\Sigma}_{21} + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{21}\Sigma_{22}^{-2}\frac{\partial \Sigma_{22}}{\partial \beta_k} - \boldsymbol{\Sigma}_{12}\Sigma_{22}^{-1}\frac{\partial \boldsymbol{\Sigma}_{21}}{\partial \beta_k} \\
\frac{\partial \bar{\boldsymbol{\mu}}}{\partial \beta_k} &= \frac{\partial \boldsymbol{\mu}_1}{\partial \beta_k} + \frac{\partial \boldsymbol{\Sigma}_{12}}{\partial \beta_k}\Sigma_{22}^{-1}\varepsilon - \boldsymbol{\Sigma}_{12}\Sigma_{22}^{-2}\frac{\partial \Sigma_{22}}{\partial \beta_k}\varepsilon - \boldsymbol{\Sigma}_{12}\Sigma_{22}^{-1}\frac{\partial \varepsilon}{\partial \beta_k} \\
\frac{\partial \varepsilon}{\partial \beta_k} &= -\frac{\partial \mu_2}{\partial \beta_k}
\end{aligned} \tag{22}$$

As $\boldsymbol{\Sigma} = \mathbf{R_e M R_e^T}$, the derivative of sigma in terms of the mosaicity model parameters is given by

$$\frac{\partial \boldsymbol{\Sigma}}{\partial \beta_k} = \mathbf{R_e}\frac{\partial \boldsymbol{M}}{\partial \beta_\mathbf{k}}\mathbf{R^T}, \tag{23}$$

where the local reflection-specific coordinate system is approximated to be fixed during optimisation of crystal cell and orientation parameters.

The Fisher information matrix requires evaluation of the expected value of the second derivatives.

The expected values of the first and second moments of the distribution of the spots about the Ewald Sphere are $E[\epsilon_i] = 0$ and $E[\epsilon_i^2] = \Sigma_{22i}$. The first and second moments of the general impact of the spot in the local coordinate system are $E[(\bar{\mathbf{x}}_i - \bar{\boldsymbol{\mu}}_i)] = 0$ and $E[S_i] = \bar{\boldsymbol{\Sigma}}_i$. The elements of the Fisher information matrix only require the first derivatives and are calculated as

$$\begin{aligned}
I_{kl} &= \frac{1}{2}\sum_i^N c_{\mathrm{tot},i}\left[\Sigma_{22i}^{-2}\frac{\partial \Sigma_{22i}}{\partial \beta_k}\frac{\partial \Sigma_{22i}}{\partial \beta_l} + 2\Sigma_{22i}^{-1}\frac{\partial \epsilon_i}{\partial \beta_k}\frac{\partial \epsilon_i}{\partial \beta_l}\right] \\
&+ \frac{1}{2}\sum_i^N c_{\mathrm{tot},i}\left[\mathrm{tr}\left(\bar{\boldsymbol{\Sigma}}_i^{-1}\frac{\partial \bar{\boldsymbol{\Sigma}}_i}{\partial \beta_k}\bar{\boldsymbol{\Sigma}}_i^{-1}\frac{\partial \bar{\boldsymbol{\Sigma}}_i}{\partial \beta_l} + 2\bar{\boldsymbol{\Sigma}}_i^{-1}\left(\frac{\partial \bar{\boldsymbol{\mu}}_i}{\partial \beta_k}\frac{\partial \bar{\boldsymbol{\mu}}_i}{\partial \beta_l}\right)\right)\right]
\end{aligned} \tag{24}$$

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